

Cellular Systems

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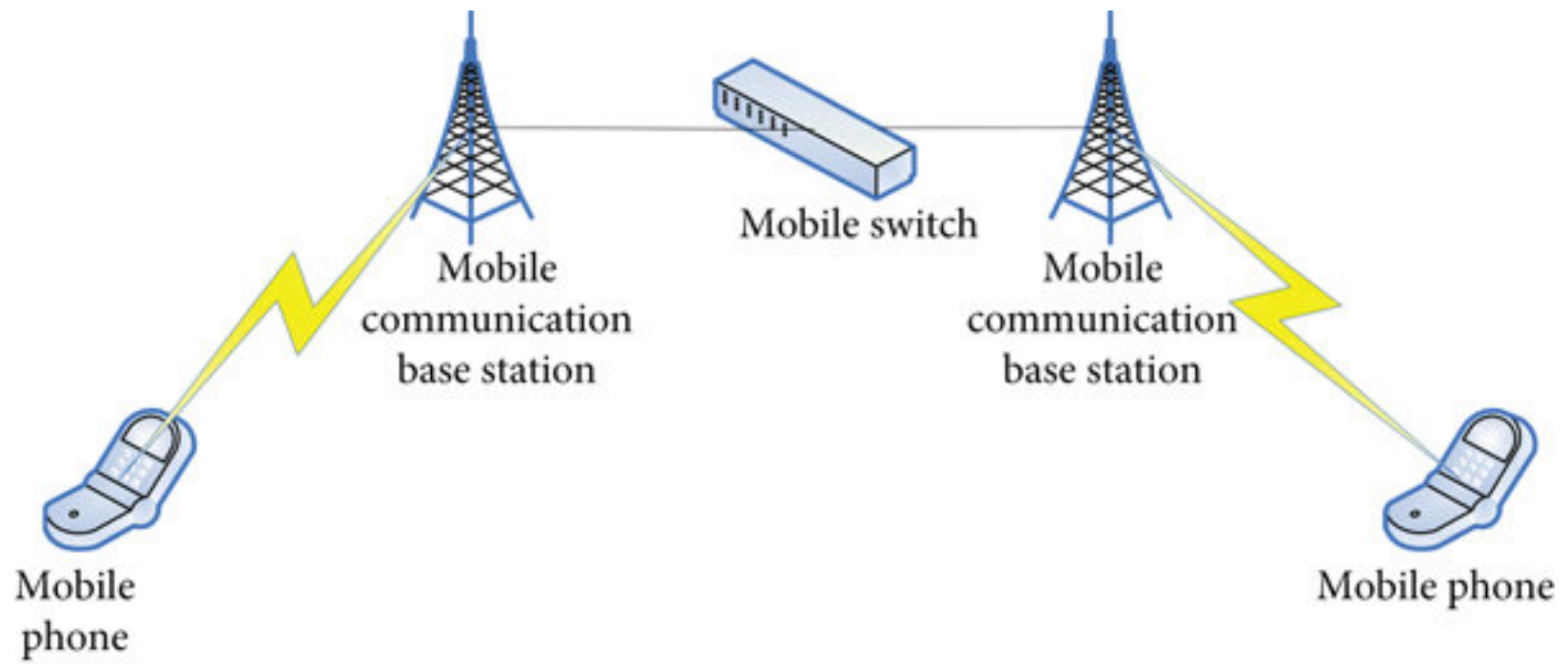
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How mobile networks work





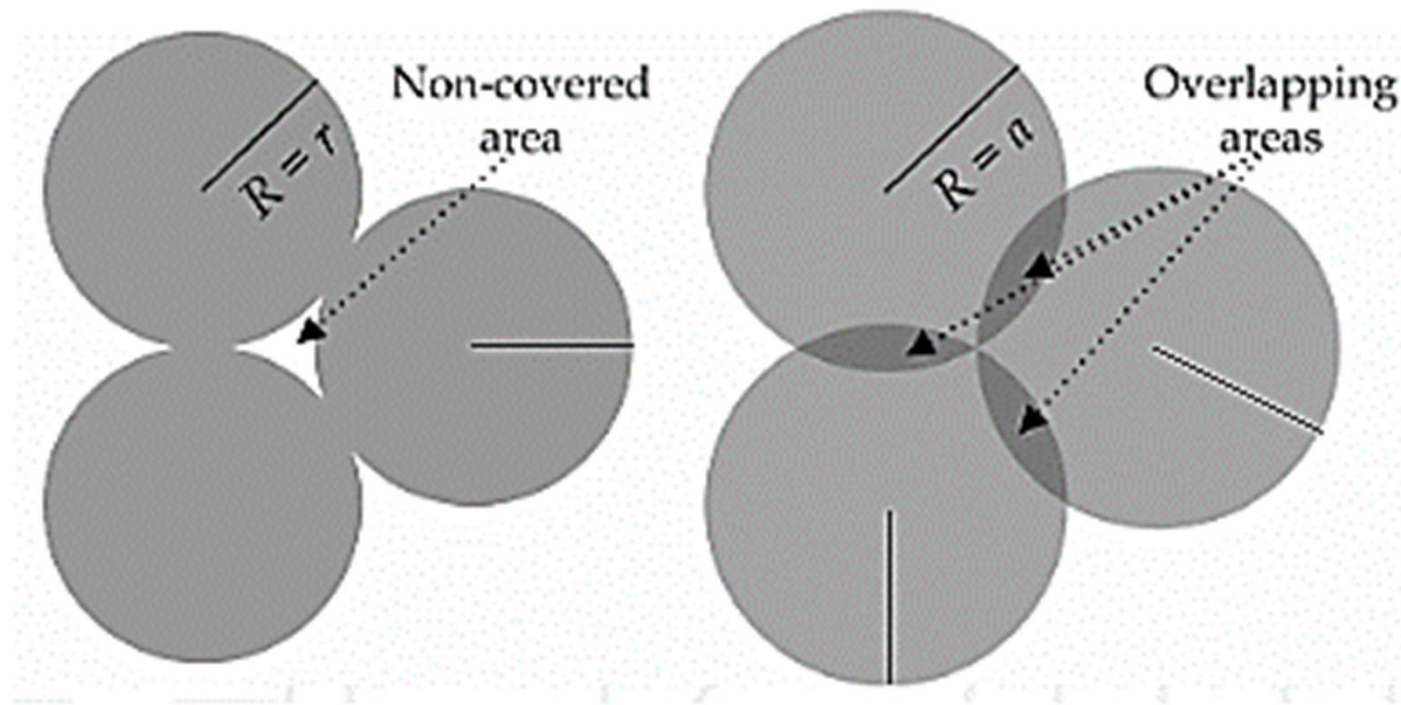
Goals

- Low power transmitter system
- Increase network capacity
- Frequency reuse
- Build robust scaleable system
- Architecture to deal with different user densities at different places

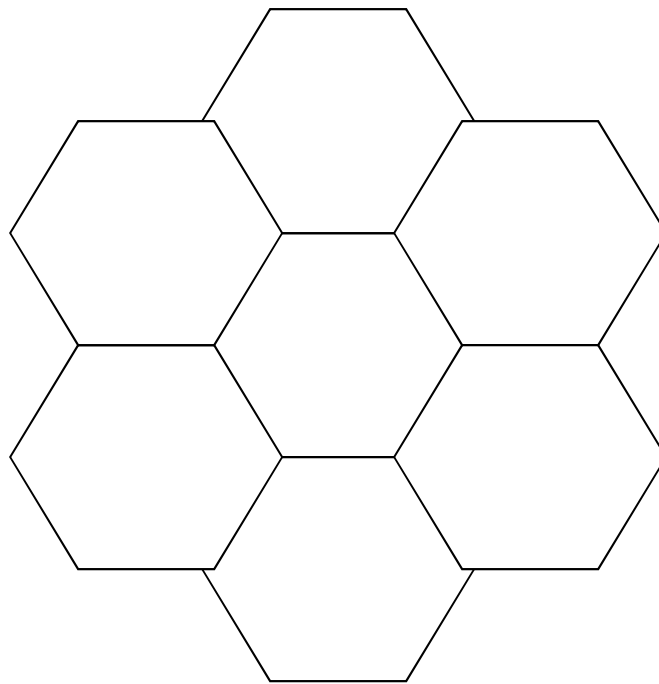
Idea!

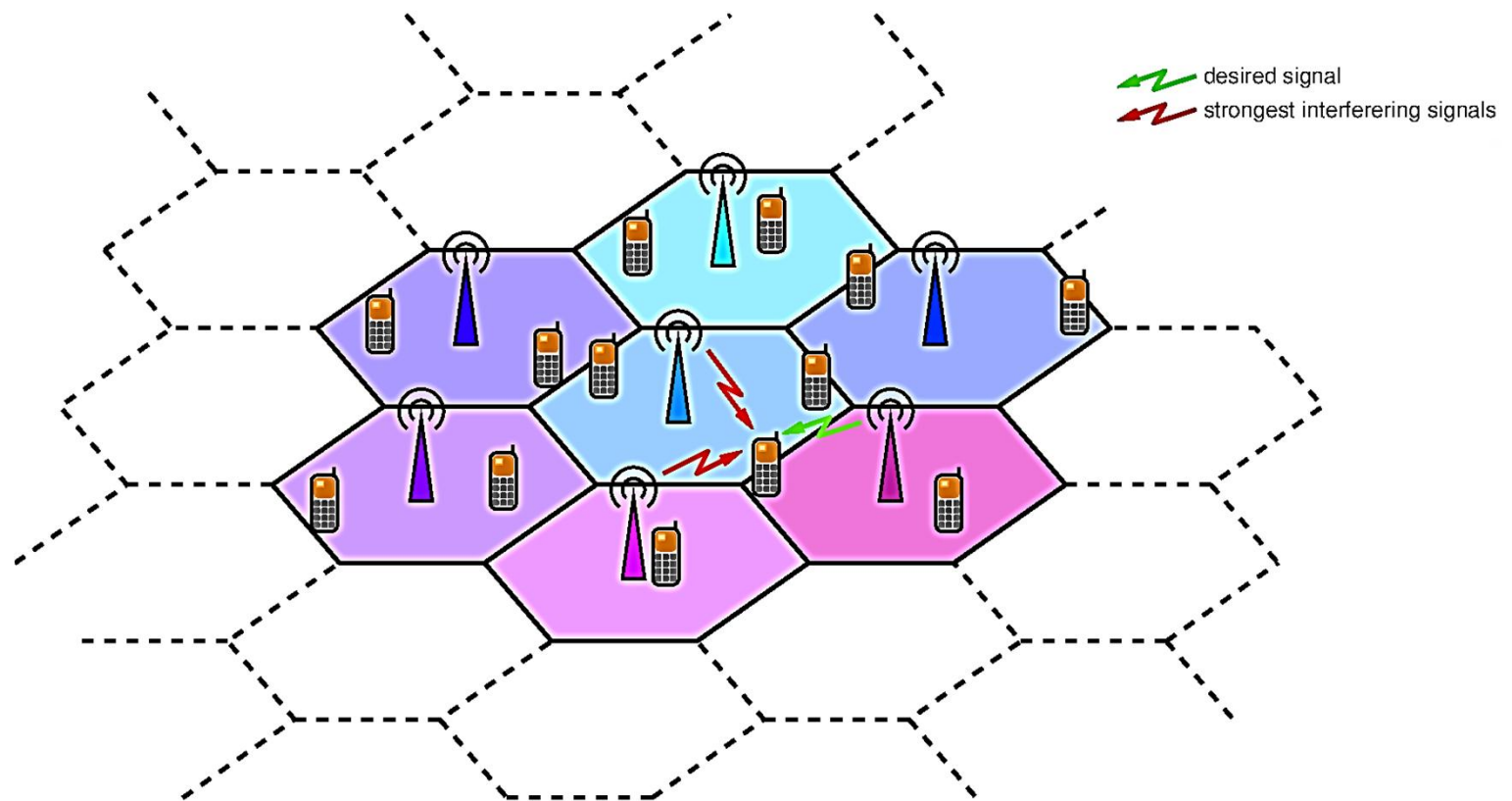
- Partition the region into smaller regions called cells.
- Each cell gets at least one base station or tower
- Users within a cell talk to the tower
- How can we divide the region into cells?

Cell structure



“Cell”ular Structure





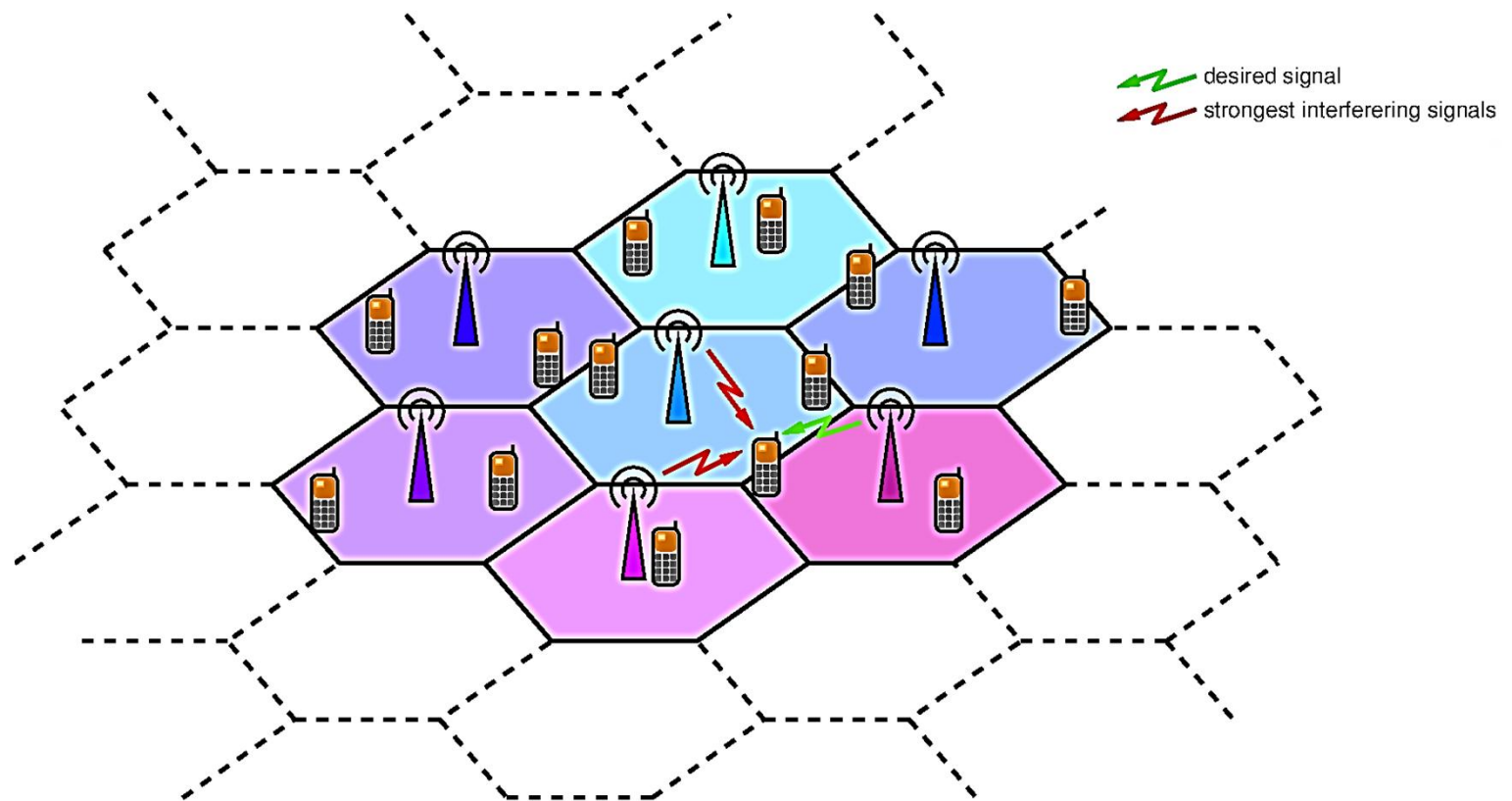
Properties of Cell structure

- Typical Cell sizes
 - some cites few hundred meters
 - country side few tens of kilometers
- Advantages of cell structures:
 - more capacity due to frequency reusage
 - less transmission power needed
 - more robust, tolerate failures
 - deals interference, transmission area locally
- Problems:
 - fixed network needed for the base stations
 - handover (changing from one cell to another) necessary
 - interference with other cells

Inside a cell

- Center-excited cell where the tower is placed somewhat near the center with a omni-directional antenna
- Edge-excited cell where the towers are placed on three of the six corners with sectored directional antennas.

Cell reuse is the fundamental concept that allows the same set of radio frequencies (channels) to be used repeatedly in different geographical areas, provided that the cells using the same frequencies are sufficiently far apart to avoid interference. This greatly increases the capacity of a cellular network.



Cell Reuse/ Frequency Reuse

- Cell structure can reuse frequency only when certain distance is maintained between cells that use the same channels.
- Fixed frequency assignment:
 - certain frequencies are assigned to a certain cell
 - problem: different traffic load in different cells
- Dynamic frequency assignment:
 - base station chooses frequencies depending on the frequencies already used in neighbor cells
 - more capacity in cells with more traffic
 - assignment can also be based on interference measurements

Interference

- Co-channel interference
 - Signals from cells that share a channel cause co-channel interference
 - Can't remove it by increasing power.
- Adjacent channel interference
 - Signals from adjacent cells cause this.
 - Use filter to reduce it
- But, available channels decrease for incoming calls.

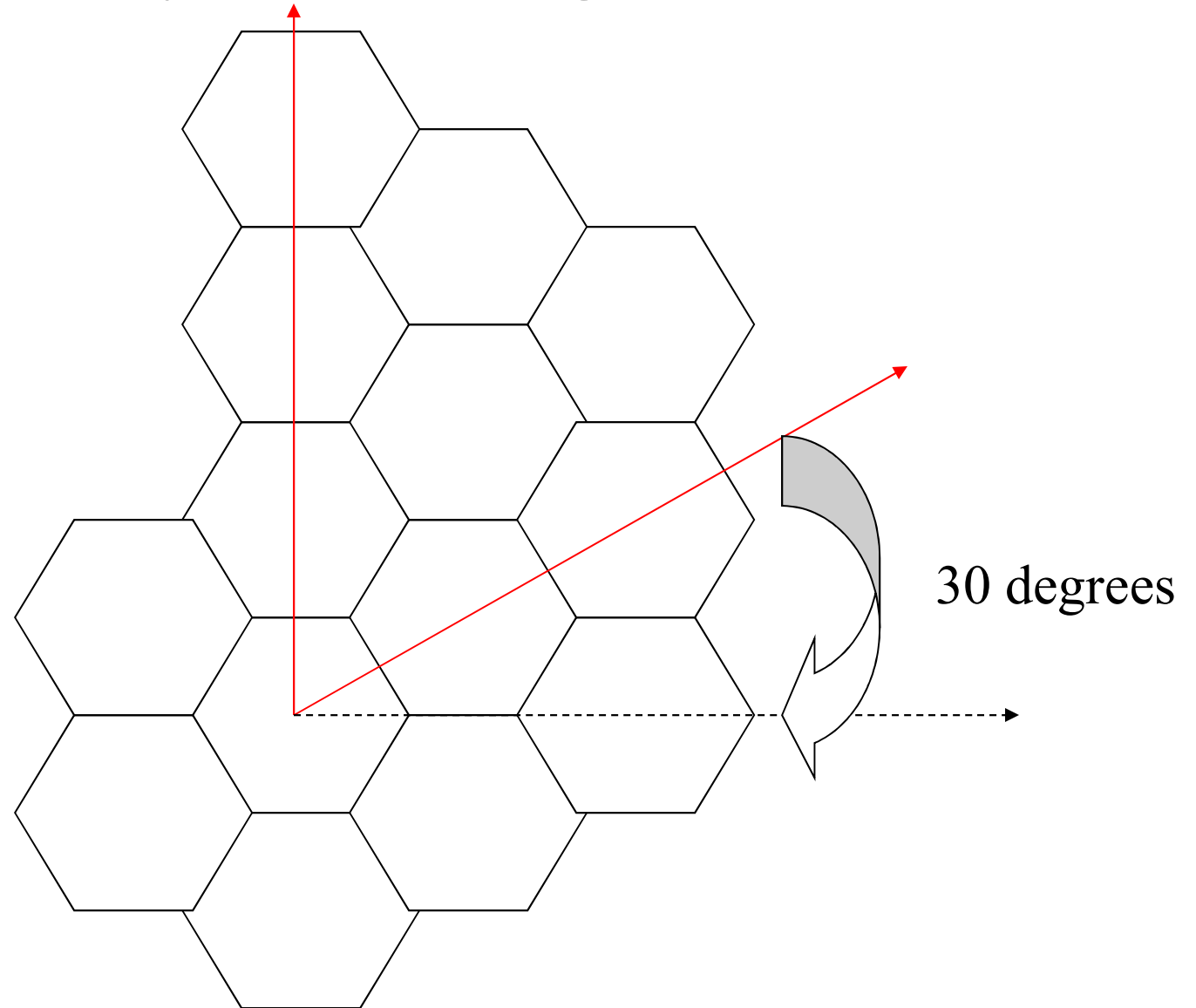
Frequency reuse factor and Capacity

- Total available channels = S
- N “adjacent” cells (called a cluster) share S channels
- System has M clusters
- Each cell gets k channels

$$S = k N$$

- Capacity of the system is $C = MkN$
- Frequency reuse factor is $1/ N$

Geometry of Hexagonal Cell



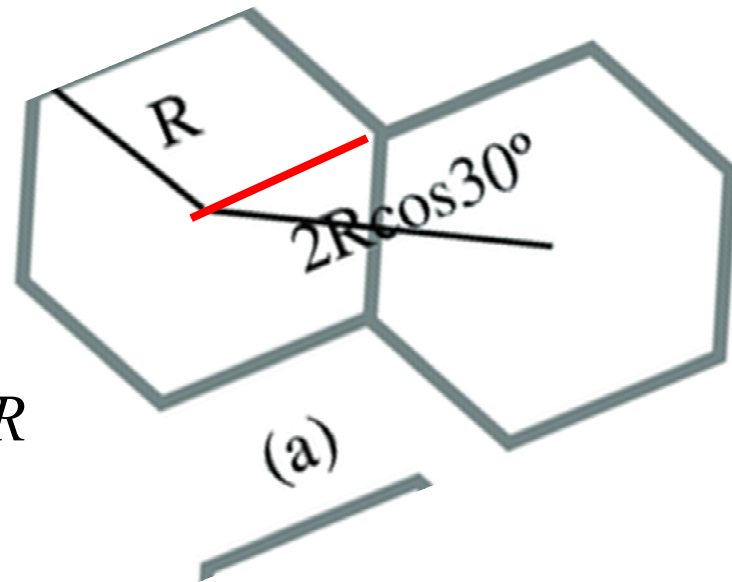
Distance calculation

- $(u1,v1)$ and $(u2,v2)$ are centers of two cells
- Distance D
- Radius is R for a cell.

$$b = R \cos 30$$

$$AB = 2b = 2R \cos 30$$

$$AB = 2 \frac{\sqrt{3}}{2} R = \sqrt{3} R = 1.732 R$$



- Distance between adjacent cells is $1.732 R$

Co-channel Interference and System Capacity

- **Co-Channel Cells:** Cells that use same frequency
- **Co-Channel Interference:** Interference between signals from these cells. It **cannot be combated by simply increasing the carrier power of a transmitter.**
- An increase in carrier transmit power increases the interference to neighboring co-channel cells.
- To reduce co-channel interference, **co-channel cells must be physically separated by a minimum distance** to provide sufficient isolation due to propagation

- The **co-channel interference ratio** is independent of the **transmitted power** and becomes a function of the radius of the cell (R) and the distance between centers of the nearest co-channel cells (D).
- By increasing the ratio of D/R , *the spatial separation between co-channel* cells relative to the coverage distance of a cell is increased.
- Thus, interference is reduced from improved isolation of RF energy from the co-channel cell. The parameter Q , called the *co-channel reuse ratio*, is related to the cluster size

$$Q = \frac{D}{R} = \sqrt{3N}$$

- A small value of Q provides larger capacity since the cluster size N is small.
- A large value of Q improves the transmission quality, due to a smaller level of co-channel interference.
- A trade-off must be made between these two objectives in actual cellular design.

	Cluster Size (N)	Co-channel Reuse Ratio (Q)
$i = 1, j = 1$	3	3
$i = 1, j = 2$	7	4.58
$i = 2, j = 2$	12	6
$i = 1, j = 3$	13	6.24

- Let i_0 be the number of co-channel interfering cells. Then, the signal-to interference ratio (S/I or SIR) for a mobile receiver which monitors a forward channel can be expressed as:

$$\frac{S}{I} = \frac{S}{\sum i_0}$$

- where S is the desired signal power from the desired base station and I_i is the interference power caused by the i^{th} interfering co-channel cell base station.

- The average received power P_r at a distance d from the transmitting antenna is approximated by

$$P_r = P_0 \left(\frac{d_0}{d} \right)^n$$

$$P_r(\text{dBm}) = P_0(\text{dBm}) - 10n \log_{10} \left(\frac{d}{d_0} \right)$$

- where P_0 is the power received at a close-in reference point in the far field region of the antenna at a small distance d_0 from the transmitting antenna and n is the path loss exponent.

- Let D_i is the distance of the i^{th} interferer from the mobile, the received power at a given mobile due to the i^{th} interfering cell will be proportional to $(D_i)^{-n}$
- Let the transmit power of each base station is equal and the path loss exponent is the same throughout the coverage area.

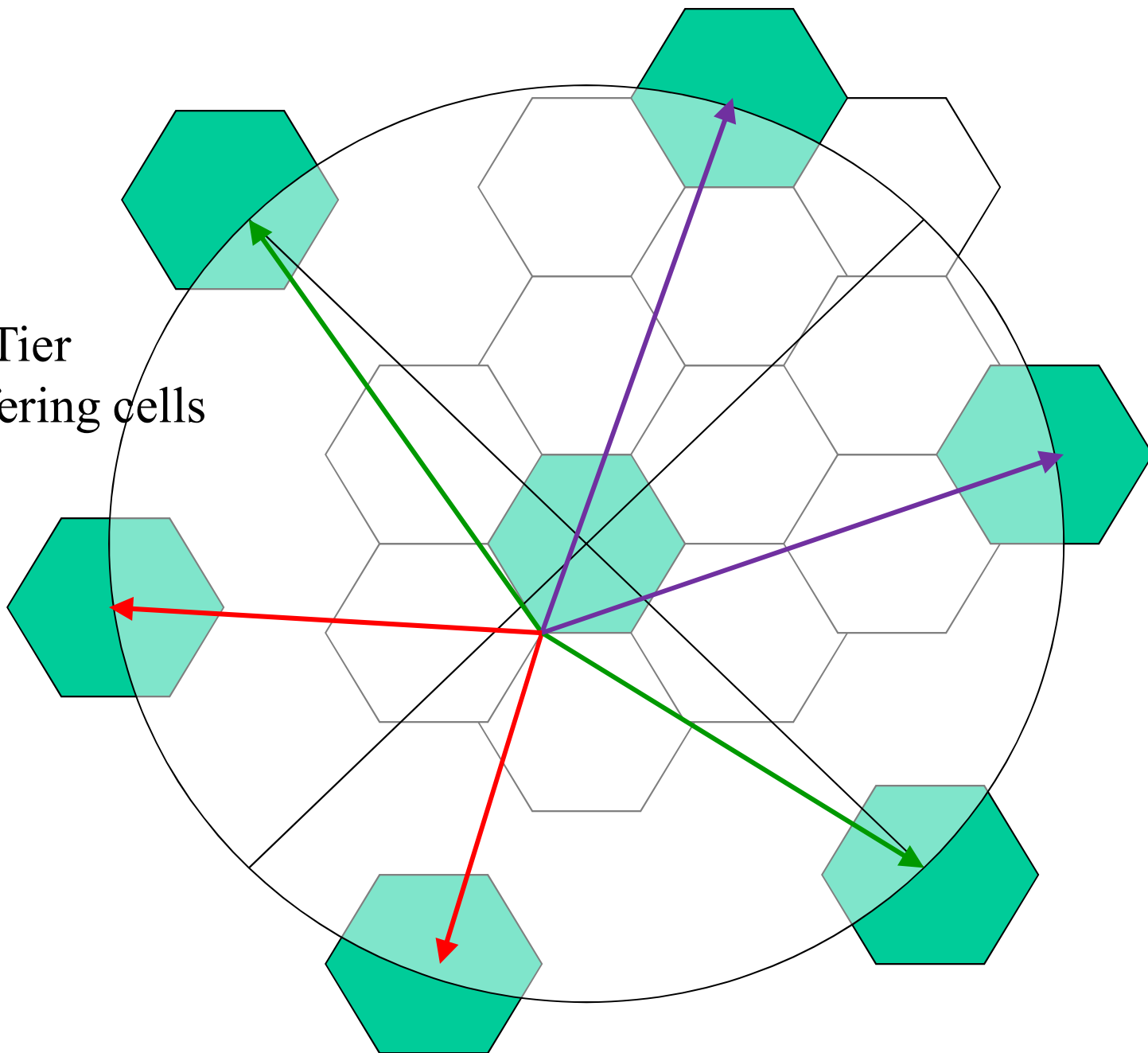
S/I for a mobile can be approximated as

$$\frac{S}{I} = \frac{R^{-n}}{\sum_{i=1} (D_i)^{-n}}$$

(The path loss exponent typically ranges between 2 and 4 in urban cellular systems.)

$$\frac{S}{I} = \frac{(D/R)^n}{i_0} = \frac{(\sqrt{3N})^n}{i_0}$$

First Tier
Interfering cells



Signal-to-Interference ratio for the worst case

$$\frac{S}{I} = \frac{R^{-4}}{2(D - R)^{-4} + 2(D + R)^{-4}} \quad (\text{Assuming } n=4)$$

$$\frac{S}{I} = \frac{1}{2(O - 1)^{-4} + 2(O + 1)^{-4}}$$

Example

If a signal-to-interference ratio of 15 dB is required for satisfactory forward channel performance of a cellular system, what is the frequency reuse ratio and cluster size that should be used for maximum capacity if the path loss exponent is (a) $n = 4$, (b) $n = 3$?

- Assume that there are six co-channel cells in the first tier, and all of them are at the same distance from the mobile. Use suitable approximations.

Solution

$$(a) \ n = 4 \qquad \frac{S}{I} = \frac{(D/R)^n}{i_0} = \frac{(\sqrt{3N})^n}{i_0}$$

- First, let us consider a seven-cell reuse pattern.
Then the co-channel reuse ratio

$$D/R = 4.58$$

- The signal-to-noise interference ratio is given by

$$S/I = (1/6) \times (4.58)^4 = 75.3 = 18.66 \text{ dB}$$

- Since this is greater than the minimum required S/I, N = 7 can be used.

(b) $n = 3$

First, let us consider a seven-cell reuse pattern.

Using Equation (3.9), the signal-to-interference ratio is given by

$$S/I = (1/6) \times (4.583)^3 = 16.04 = 12.05 \text{ dB}$$

Since this is less than the minimum required S/I , we need to use a larger N .

Using Equation (3.3), the next possible value of N is 12, ($i = j = 2$).

The corresponding co-channel ratio is given by Equation (3.4) as

$$D/R = 6.0$$

Using Equation (3.3), the signal-to-interference ratio is given by

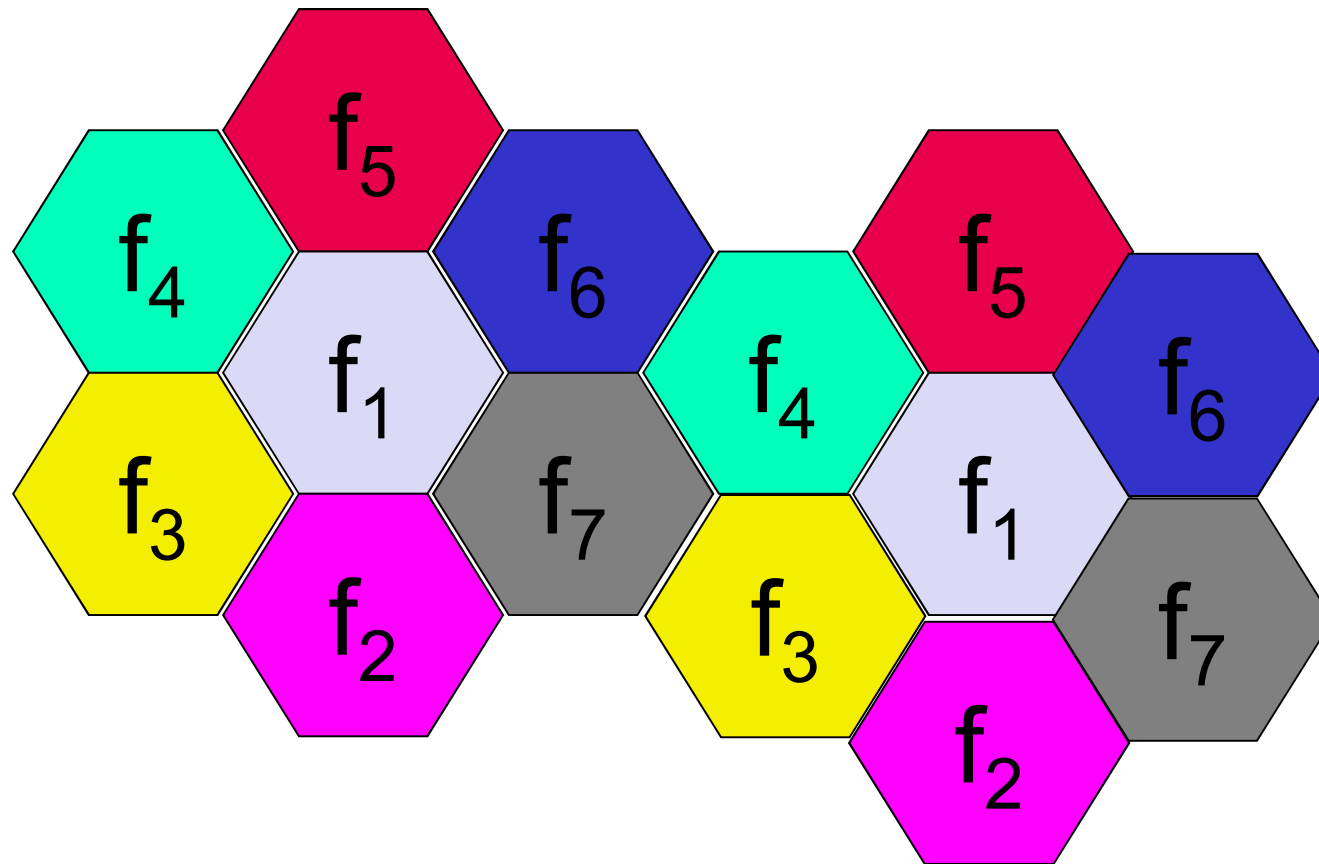
$$S/I = (1/6) \times (6)^3 = 36 = 15.56 \text{ dB}$$

Since this is greater than the minimum required S/I , $N = 12$ is used.

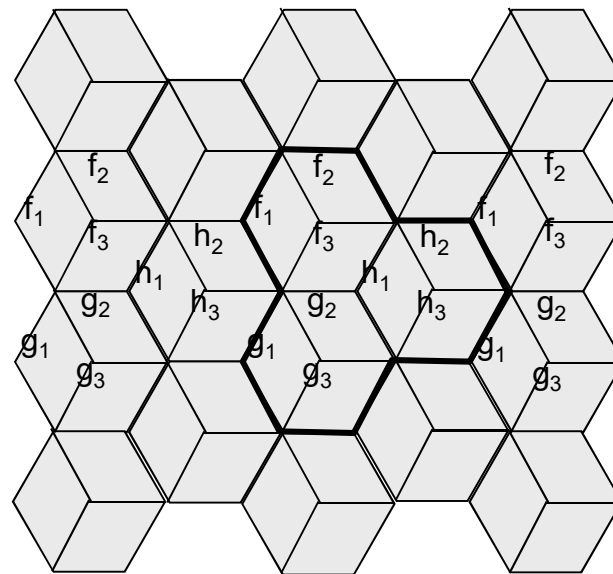
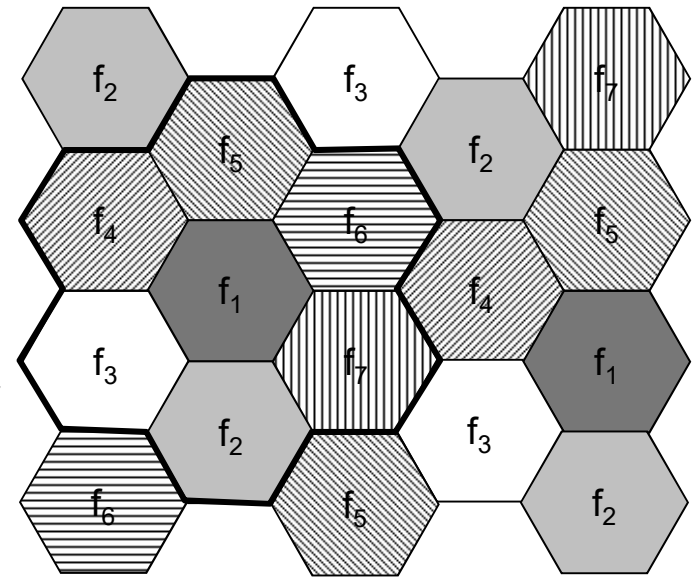
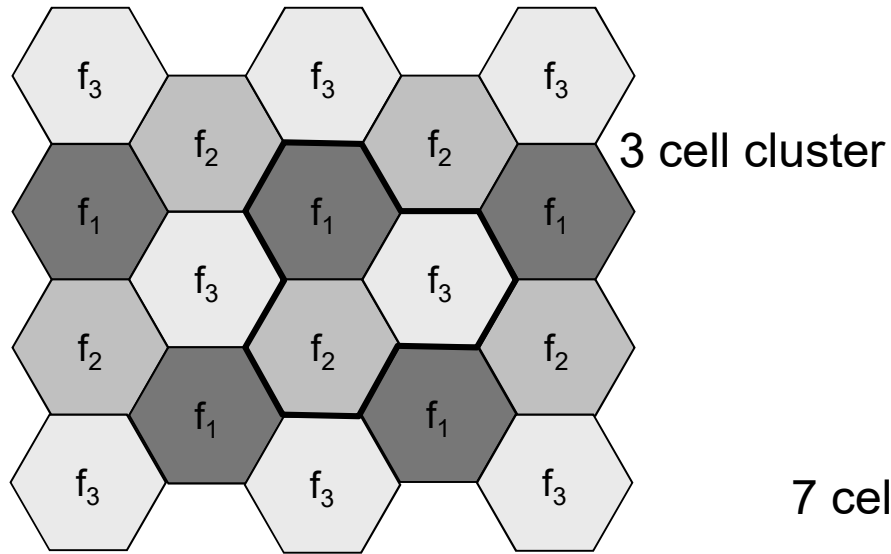
Cell reuse ratio vs Mean S/I

Cluster Size N	$Q = D/R$	Voice Channels per cell	Calls per Cell per Hour	Mean S/I dB
4	3.5	99	2610	14.0
7	4.6	56	1376	18.7
12	6.0	33	739	23.3

Standard 7 cells sharing system ($N = 7$)

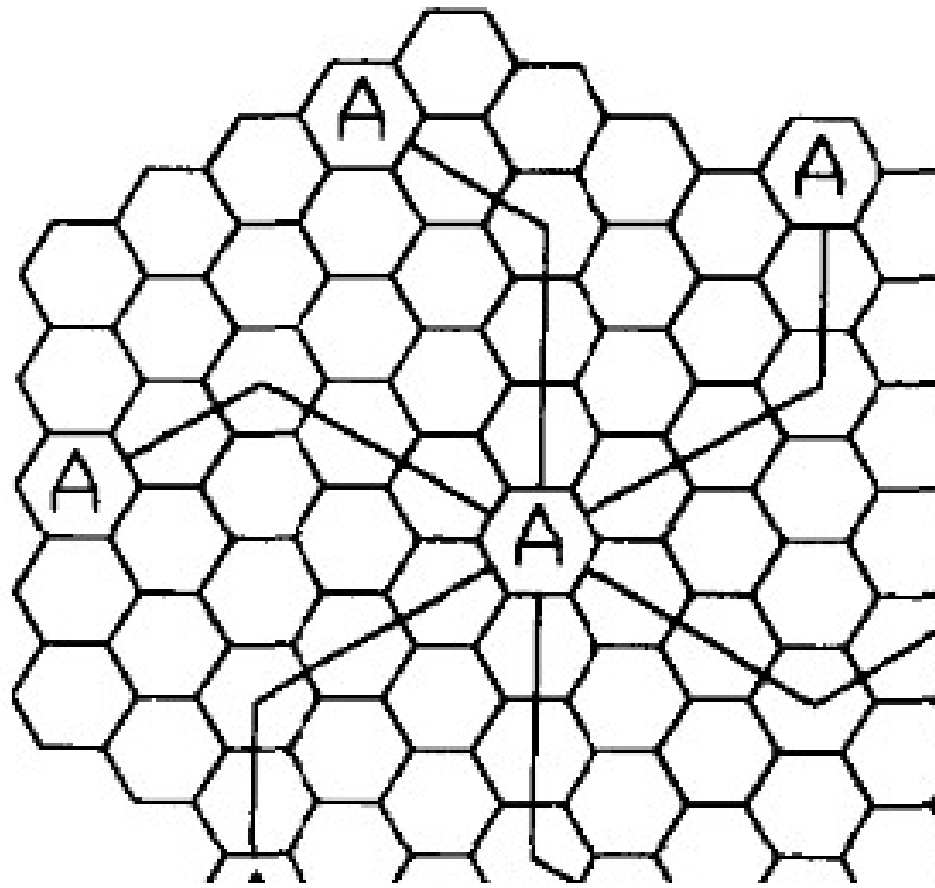


Other Common Channel Sharing



3 cell cluster
with 3 sector antennas

A bigger cluster



Cluster Size?

19

Handoff

- What happens when a user is mobile?
 - Especially when crossing a cell boundary while continuing the call.
- Handoff strategy is invoked.
 - Find a new base station
 - Process handoff
 - higher priority over new call invocation

$$\Delta = P_{r \text{ handoff}} - P_{r \text{ minimum}}$$

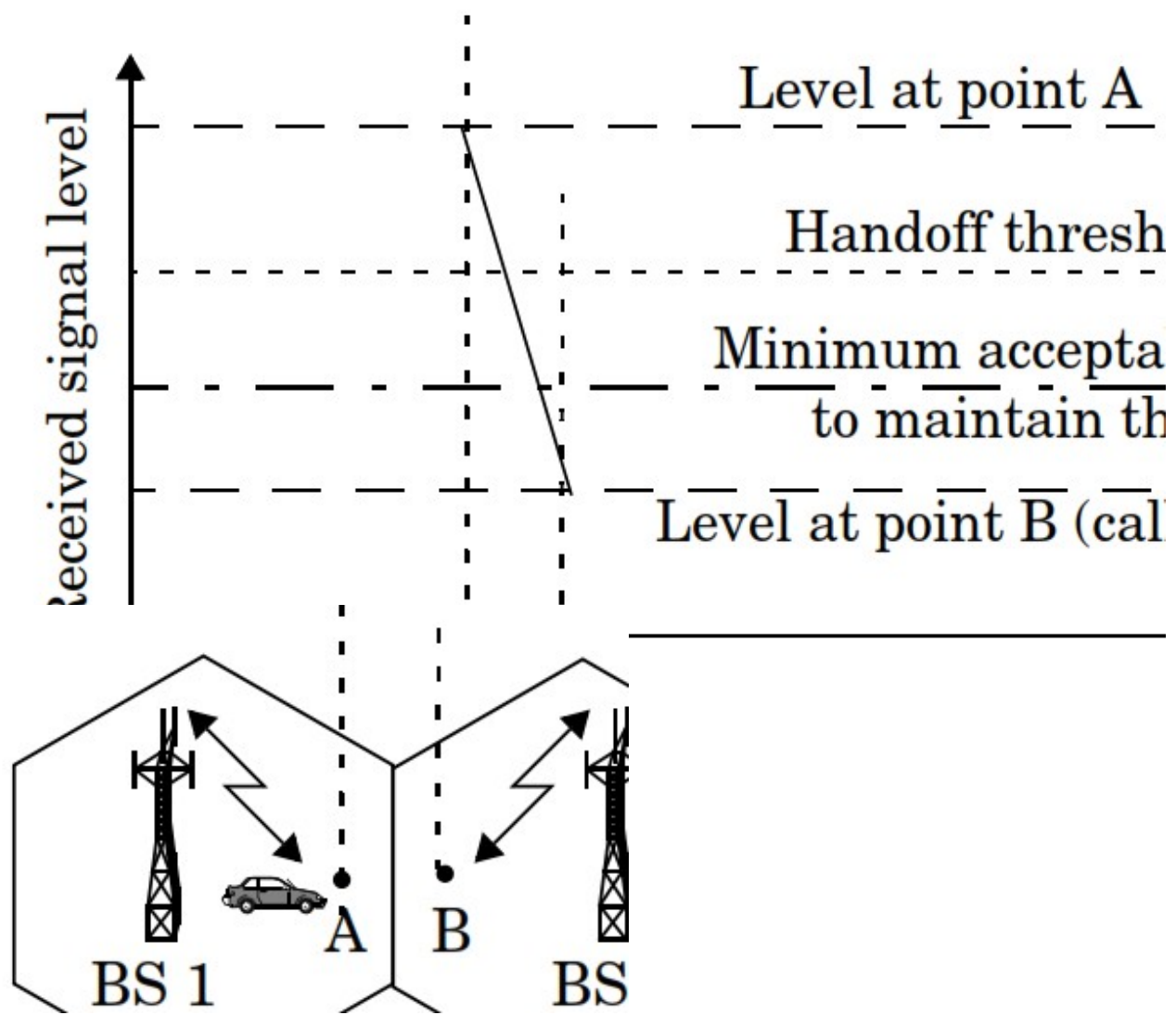
Who and When

- Who initiates handoff
 - Network directed (tower determines)
 - Terminal assisted (user helps the tower)
 - Terminal directed (user determines)
- When to initiate handoff
 - When the mean signal (over some predetermined time) is below some threshold

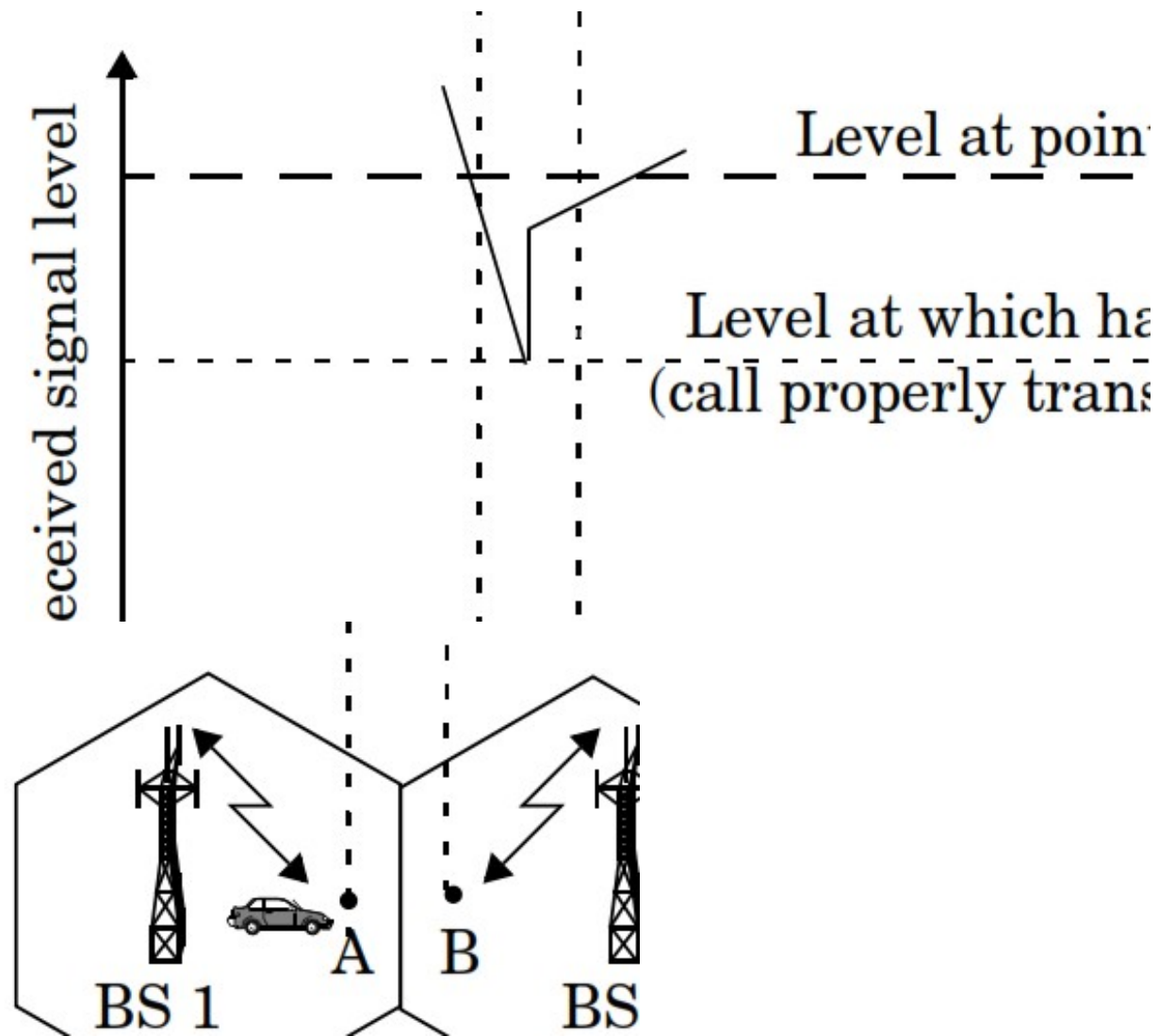
Types of Handoff

- Hard handoff
 - Mobile user is **passed between disjoint towers that assign different frequency** or adapt different air-interface technology
- Soft handoff
 - Mobile user **communicates to two towers simultaneously** and the signal is treated as a multipath signal

(a) Improper handoff situation

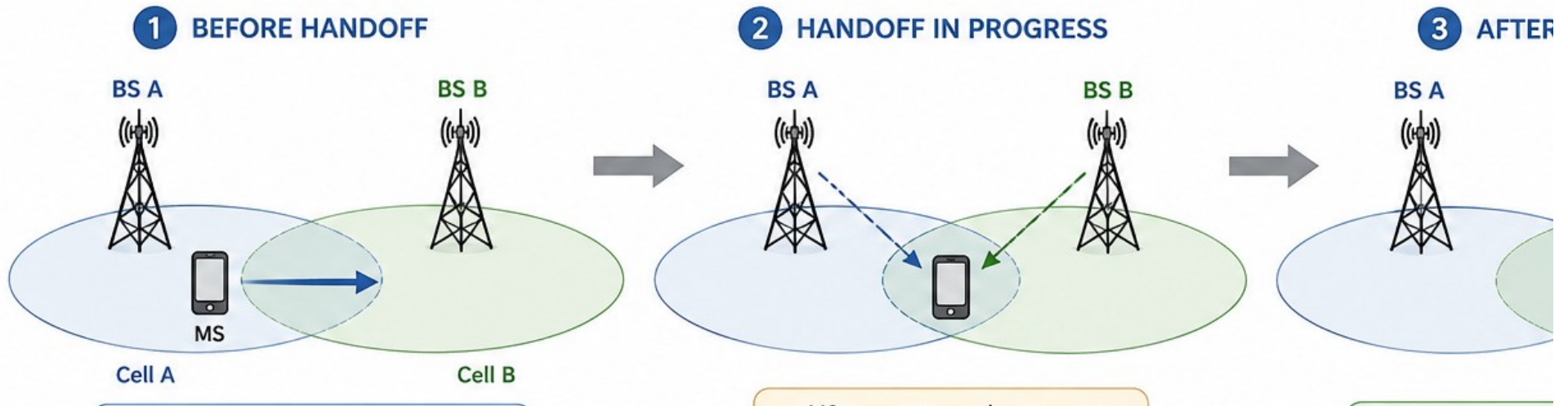


(b) Proper
handoff situation



HANDOFF IN MOBILE COMMUNICATION

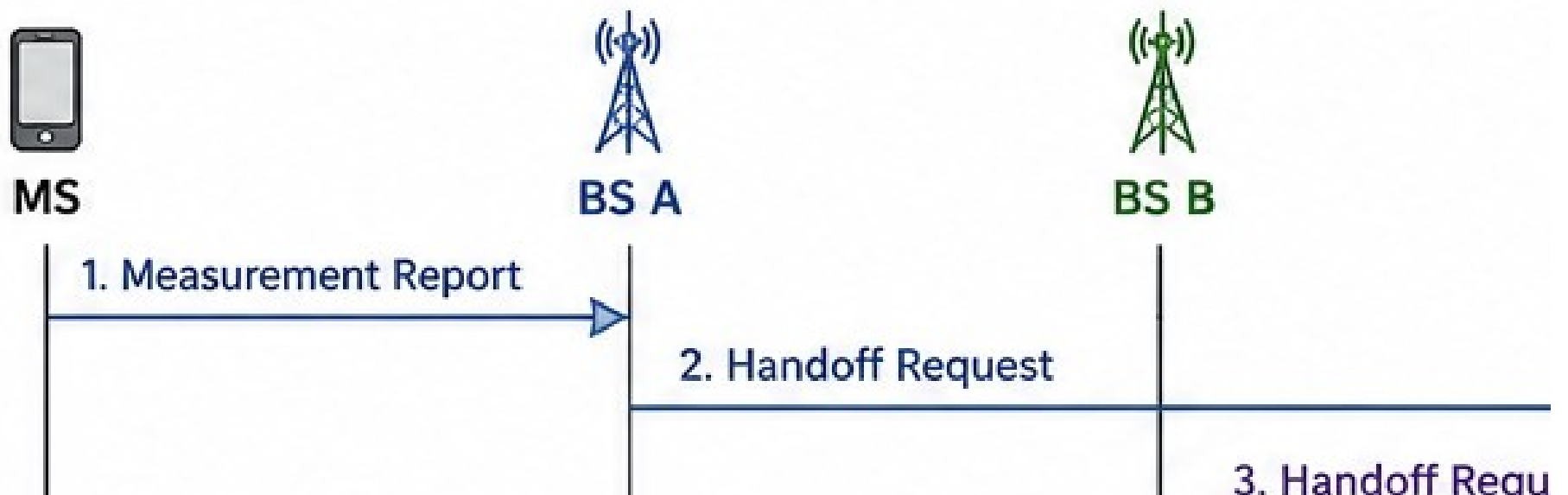
Handoff is the process of transferring an ongoing call/data session of a mobile user (MS) from one cell (labeled Cell A) to another cell without disrupting the connection.



MS is in coverage of Cell A.
Communication is with BS A.

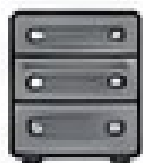
IV.
MS r
H.

HANDOFF SIGNALING FLOW (EXAMPLE)

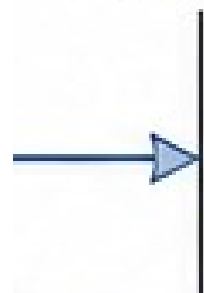


oves to overlap area.
asures signals and detects
that BS B is better.
off decision is made.

Connecti
Call/Data c



**MSC / Core
Network**



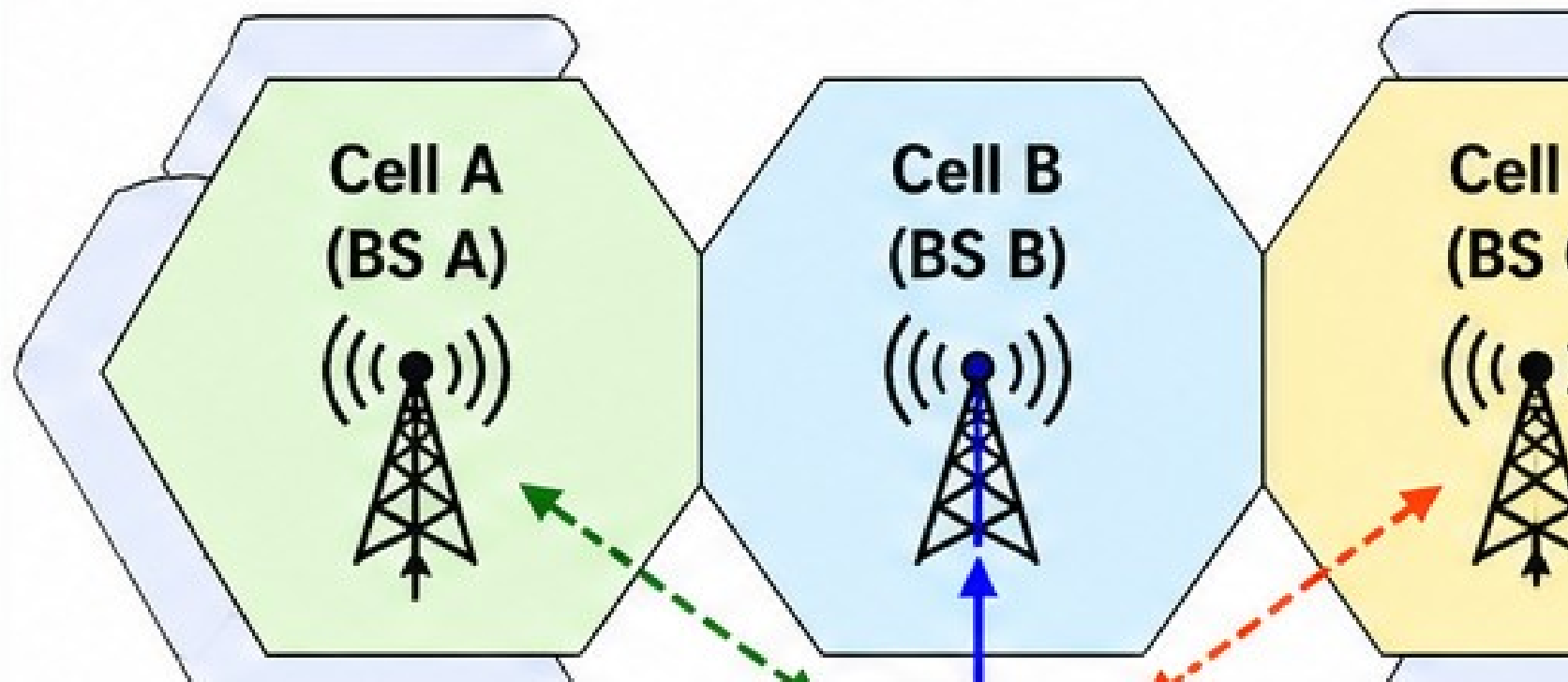
HOW HANDOFF

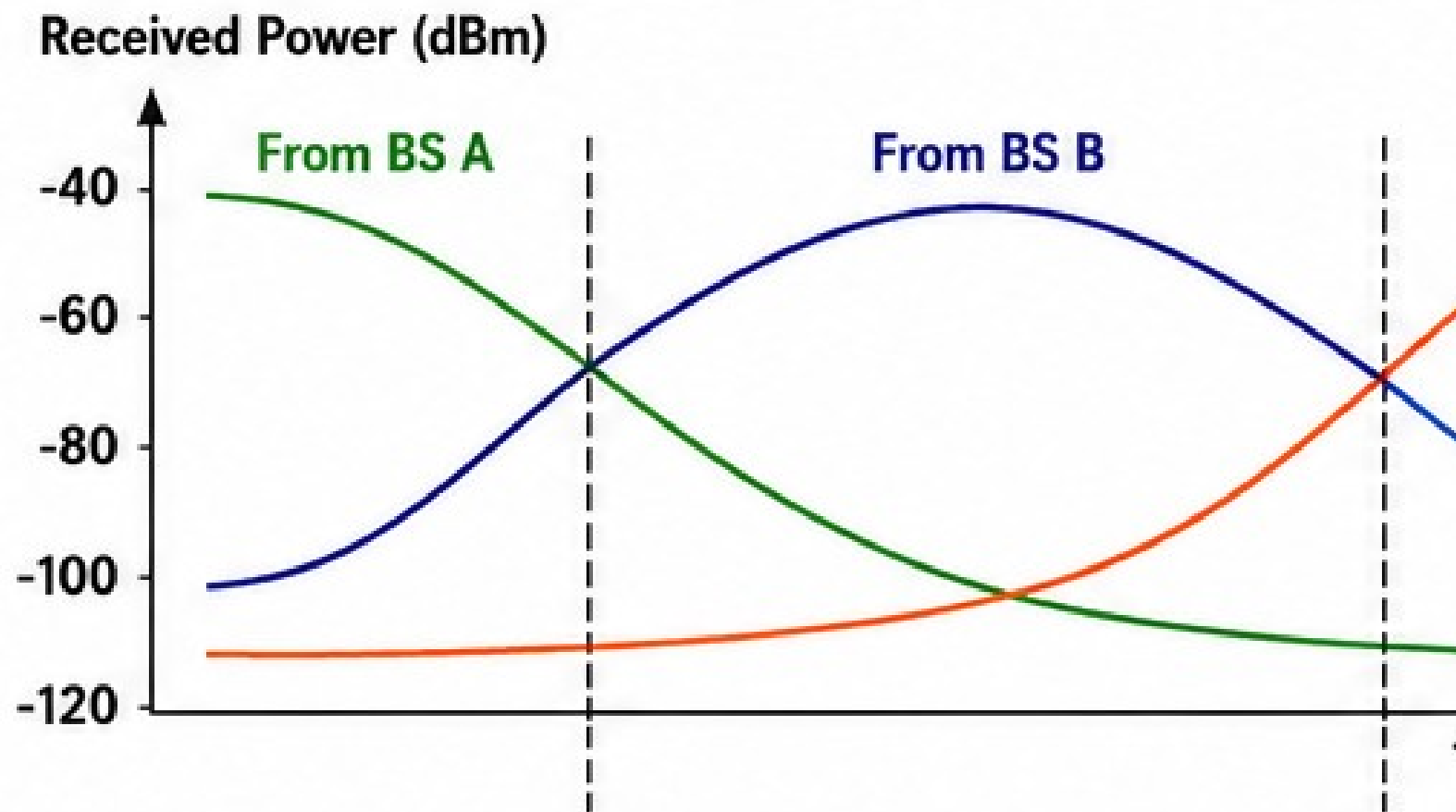
- 1 MS measures signal strength of su
sends Measurement Report to ser
- 2 BS A decides to handoff and sends
to MSC/Core Network.
- 3 MSC checks BS B availability and s
- 4 MSC allocates resources in BS B a
to BS A.

MAHO

- Stands for **Mobile Assisted Handoff**.
- Mobile Assisted Handoff (MAHO) is a technique in which the **mobile devices assist the (BSC) to transfer a call** to another BSC.
- It is used in **GSM cellular networks**. In other systems, like **AMPS**, a handoff is solely the job of the BSC and the Mobile Switching Centre (MSC), without any participation of the mobile device.
- In GSM, when a mobile station is not using its time slots for communicating, **it measures signal quality to nearby BSC and sends this information to the BSC**. The BSC performs handoff according to this information.

In MAHO, the mobile station (MS) measures the signal strength from surrounding base stations and decides when to request a handoff.





Mobile decides to handoff
from A $\rightarrow$ B when
 $P(B) > P(A)$ by a margin H

Mobile decides to
handoff from B $\rightarrow$ C
when $P(C) > P(B)$ by a

Problems with handoff

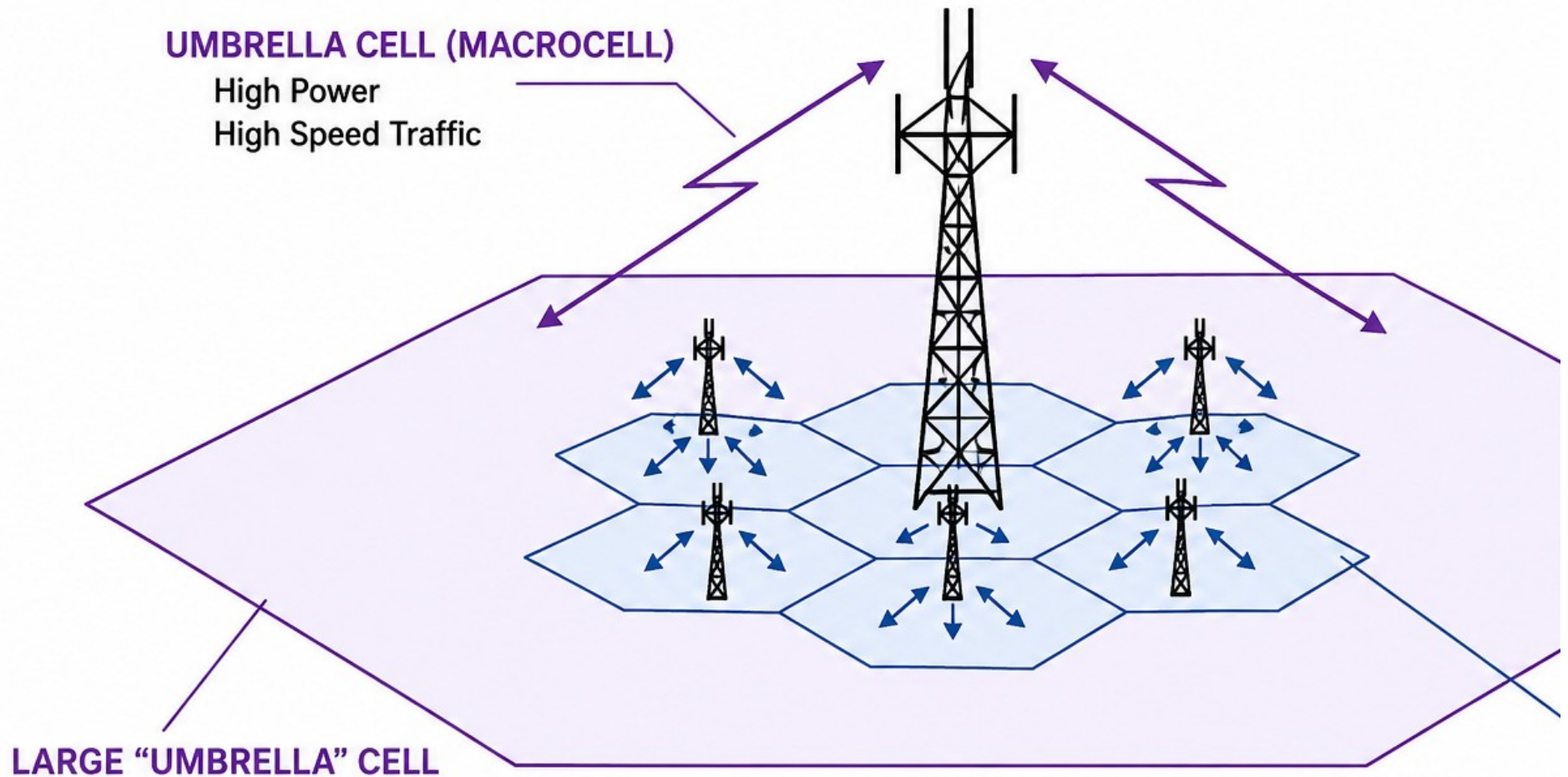
- High speed vehicles can **cross many “small” cells in a short time.**
 - **Umbrella cell.** Large cell with a powerful tower to handle high speed vehicles
- Another problem is called **cell dragging.**
 - Happens when the **user moves slowly away from the cell and the tower didn't recognize it** due to strong average signal.

UMBRELLA CELL CONCEPT

A large high-power “umbrella” cell is overlaid on a cluster of smaller microcells. It provides wide area coverage and handles high speed traffic, while microcells handle low speed traffic within the area.

UMBRELLA CELL (MACROCELL)

High Power
High Speed Traffic



Improving Capacity

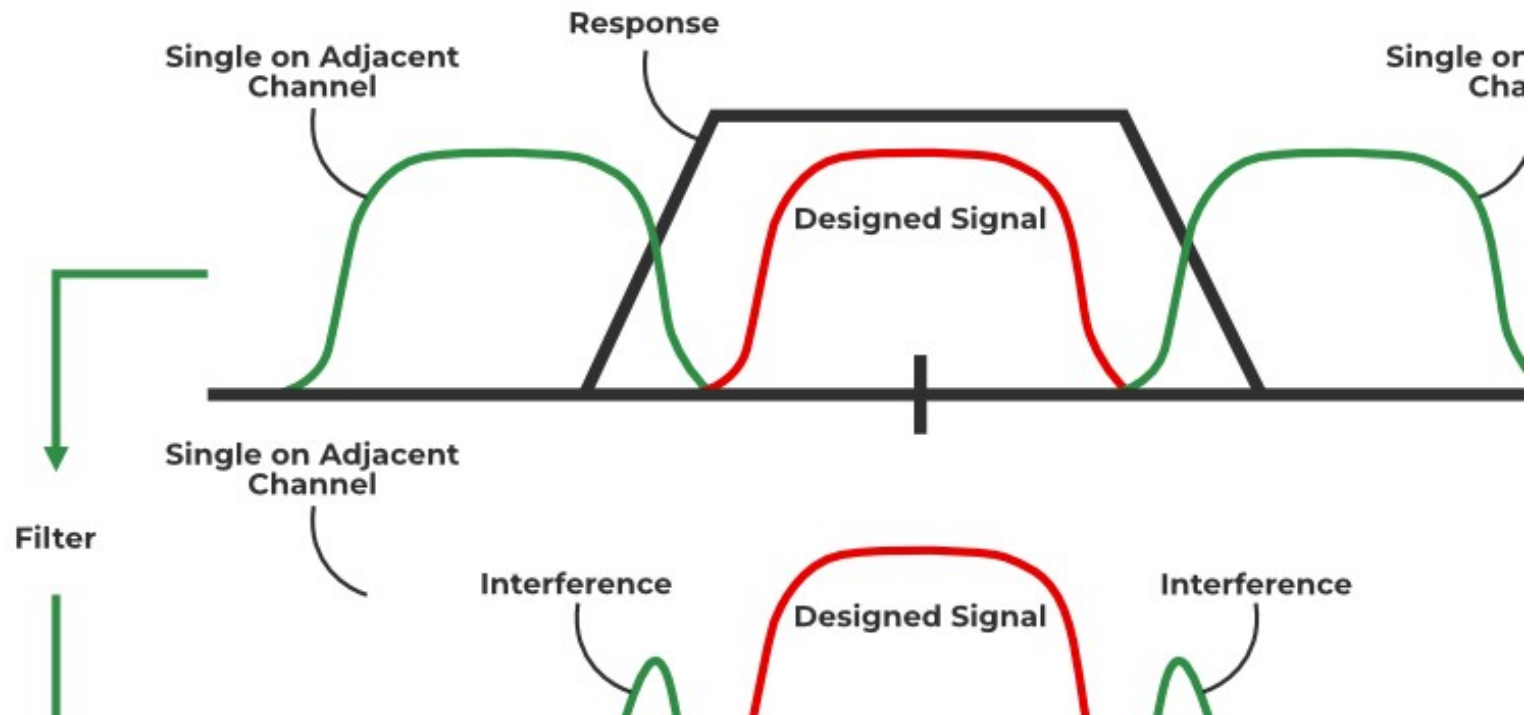
- Sectoring
- Cell splitting
 - Process of subdividing a congested cell into smaller cells.
 - Each has its own base station
 - Smaller antenna and reduced transmission power
 - These smaller cells are called microcells

Adjacent Channel Interference

It is a common problem in mobile communication systems that occurs **when signals from one channel or frequency band interfere with signals in an adjacent or nearby channel.**

This interference can degrade the quality of communication and reduce the system's overall performance.

Adjacent Channel Interference

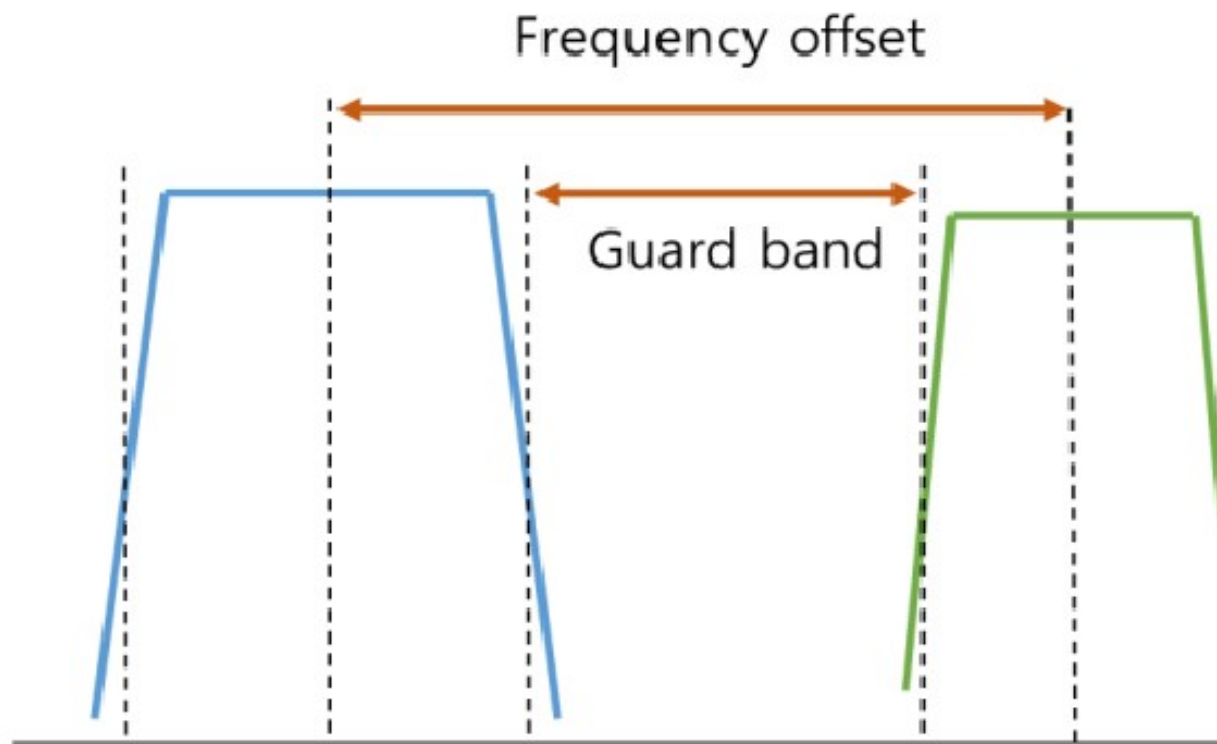


Adjacent Channel Interference

- **Frequency Reuse:** Mobile communication systems, use a technique called frequency reuse to maximize spectrum efficiency.
- In frequency reuse, the available frequency spectrum is divided into smaller channels or frequency bands, and these channels are reused in different cells within the network.
- However, adjacent cells often use frequencies that are very close to each other.

Adjacent Channel Interference

- **Guard Bands:** To prevent interference between adjacent channels, guard bands are typically allocated between channels.
- Guard bands are **unused frequency ranges that act as a buffer** to minimize interference.
- They help ensure that signals in one channel do not spill over into neighboring channels.



Causes of ACI

- **Transmitter Imperfections:** Imperfections in the transmitters or transceivers of mobile devices or base stations can lead to spurious emissions that extend into adjacent channels.
- **Non-Linear Effects:** Non-linearities in amplifiers and other components of the communication system can cause signal distortion and spectral regrowth, leading to interference in adjacent channels.
- **Frequency Drift:** Variations in the frequency of a signal source, such as a local oscillator, can cause the signal to drift into adjacent channels.
- **Receiver Selectivity:** The selectivity of the receiver, or its ability to reject signals from adjacent channels, plays a crucial role in preventing ACI.

Mitigation Techniques for ACI

- **Better Filtering:** Improved filtering at both the transmitter and receiver ends can reduce unwanted emissions into adjacent channels.
- **Frequency Planning:** Careful frequency planning and allocation of channels can help minimize interference by ensuring that neighboring cells use non-overlapping frequency bands.
- **Power Control:** Dynamic power control can be used to adjust the transmit power of mobile devices based on the proximity of neighboring cells, reducing interference.
- **Advanced Modulation Schemes:** Using advanced modulation schemes and error correction techniques can make the system more robust against interference.
- **Interference Cancellation:** Some advanced receiver techniques can be employed to cancel out unwanted adjacent channel interference.

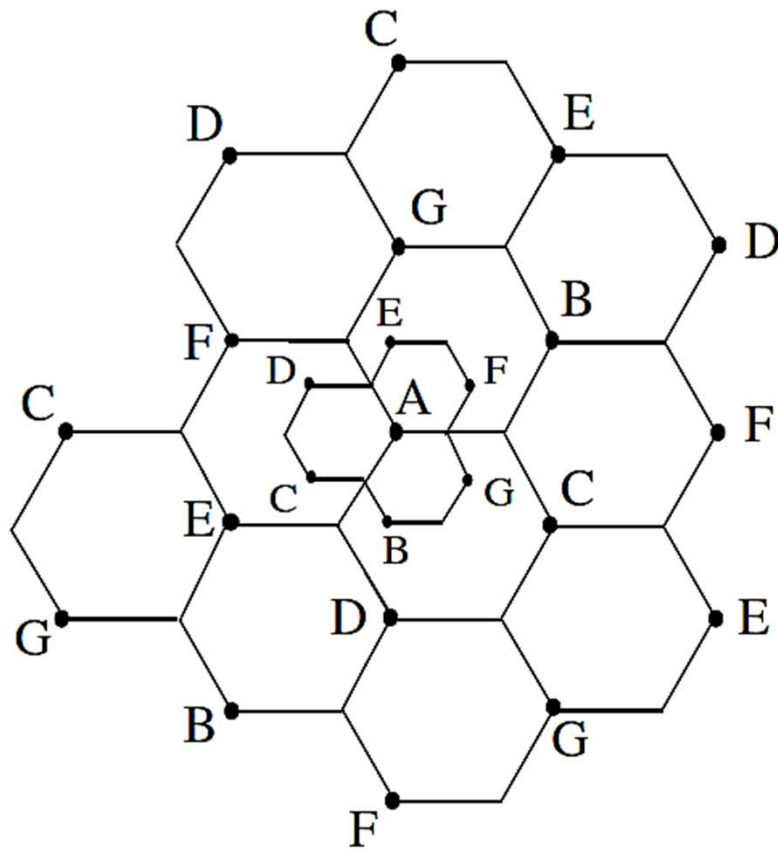
- **Regulatory Requirements:** Regulatory bodies, such as the Federal Communications Commission (FCC) in the United States, or TRAI in India, impose **limits on the amount of power that can spill over into adjacent channels** to minimize interference and ensure spectrum efficiency.
- ACI is a significant concern in mobile communication systems due to the reuse of frequency bands and the potential for signal spillage between neighboring channels. To address this issue, **careful system design, improved filtering, power control, and adherence to regulatory requirements** are essential.

Improving Coverage and Capacity in Cellular Systems

- As the demand for wireless service increases, the number of channels assigned to a cell eventually becomes insufficient to support the required number of users.
- At this point, cellular design techniques are needed to provide more channels per unit coverage area.
- Techniques such as *cell splitting*, *sectoring*, and *coverage zone approaches* are used in practice to expand the capacity of cellular systems.

Cell Splitting

- Cell splitting is the process of subdividing a congested cell into smaller cells, each with its own base station and a corresponding reduction in antenna height and transmitter power.
- Cell splitting **increases the capacity** of a cellular system since it increases the number of times that channels are reused.
- By defining new cells which have a smaller radius than the original cells and by installing these smaller cells (called *microcells*) *between the existing cells*, capacity increases due to the additional number of channels per unit area.



Base stations are placed at corners of the cells, and the area served by base station *A* is *assumed to be saturated* with traffic i.e., the blocking of base station *A* *exceeds acceptable rates*).

New base stations are therefore needed in the region to increase the number of channels in the area and to reduce the area served by the single base station.

- For the new cells to be smaller in size, the transmit power of these cells must be reduced.
- The transmit power of the new cells with radius half that of the original cells can be found by examining the received power P at the new and old cell boundaries and setting them equal to each other.
- This is necessary to ensure that the frequency reuse plan for the new microcells behaves exactly as for the original cells.

$$P_r[\text{at old cell boundary}] \propto$$

$$P_r[\text{at new cell boundary}] \propto P_{t2} ($$

where P_{t1} and P_{t2} are the transmit powers of the larger and smaller cell base stations, respectively

- Let, $n=4$

then, the received power in both the scenarios should be equal;

$$P_{t1}R^{-n} = P_{t2}(R/2)^{-n}$$

$$P_{t1}R^{-4} = P_{t2}(R/2)^{-4}$$

$$P_{t1} = P_{t2}\left(\frac{2}{R}\right)^4 R^4 = P_{t2}16$$

$$P_{t2} = \frac{P_{t1}}{16}$$

- This means, the **transmit power must be reduced by 12dB** in order to fill in the original coverage area with microcells, while maintaining the *S/I requirement*.

Advantages of Cell Splitting

- **Increased Capacity:** By reducing the coverage area of each cell, cell splitting effectively increases the network's capacity.
- **Improved Signal Quality:** Smaller cells result in shorter distances between mobile devices and base stations. This reduced distance leads to **better signal quality, reduced interference, and improved call quality.** Mobile devices in smaller cells **experience less path loss and lower signal attenuation,** which contributes to stronger and more reliable connections.

Advantages of Cell Splitting

- **Load Balancing:** In densely populated areas or during peak usage times, the cellular network operator can **redirect some users to neighboring cells with lower traffic**, distributing the user load more evenly and preventing network congestion.
- **Efficient Spectrum Utilization:** Cell splitting allows operators to reuse the same frequency channels in different cells. This reuse pattern, often referred to as "frequency reuse," maximizes spectrum utilization and minimizes interference between cells.

Limitations of Cell Splitting

- **Increased Infrastructure Costs:** For the deployment of additional base stations, antennas, and associated infrastructure
- **Handover Challenges:** Smaller cells mean more frequent handovers as mobile devices move through different cells
- **Spectrum Fragmentation:** Smaller cells reuse the same frequency bands, which can result in increased interference
- **Complexity of Interference Management:** Managing and mitigating interference through techniques like power control and advanced modulation schemes can be challenging.

Limitations of Cell Splitting

- **Increased Network Maintenance:** With more cells to manage, network maintenance and troubleshooting become more complex and labor-intensive. Identifying and resolving issues in a network with multiple cells can be time-consuming.
- **Scalability Challenges:** There is a practical limit to how small cells can become. As cells get smaller, the benefits of splitting diminish.

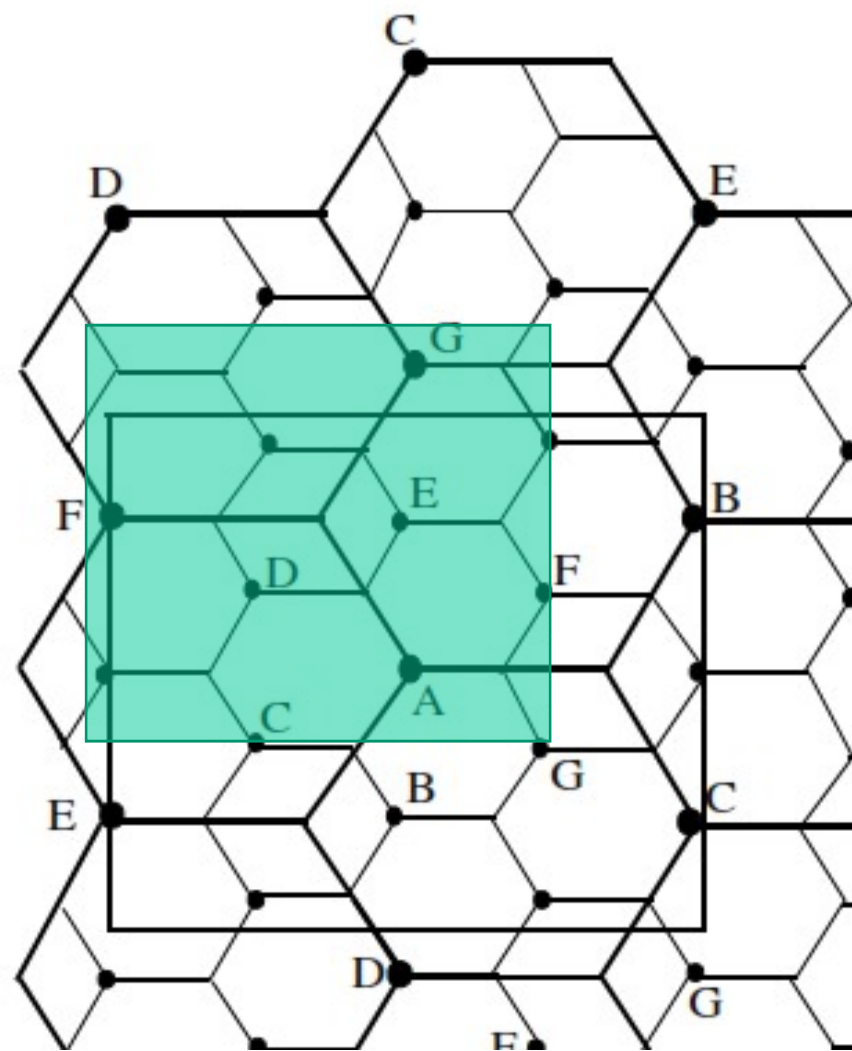
Types of Cell Splitting

There are different types of cell splitting, including:

- **Soft Cell Splitting:** This involves adjusting the transmit power of existing base stations to reduce the cell size without adding new hardware. It's a cost-effective method but may have limitations.
- **Hard Cell Splitting:** In hard cell splitting, new base stations and antennas are added to create smaller cells. This approach is more expensive but provides more significant capacity improvements.

Example

Assume each base station uses **60 channels**, regardless of cell size. If each original cell has a **radius of 1 km** and each **microcell has a radius of 0.5 km**, find the number of channels contained in a **3 km by 3 km square** centered **around A** under the following conditions: (a) without the use of microcells; (b) when the lettered microcells as shown in Figure 3.9 are used; and (c) if all the original base stations are replaced by microcells. Assume cells on the edge of the square to be contained within the square.



(a) without the use of microcells:

- A cell radius of 1 km implies that the sides of the larger hexagons are also 1 km in length.
- To cover the 3 x 3 km square centered around base station A, we need to cover 1.5 km (1.5 times the hexagon radius) toward the right, left, top, and bottom of base station A.
- This area contains **five base stations**.
- Since each base station has 60 channels, the total number of channels without cell splitting is equal to:

$$5 \times 60 = 300 \text{ channels.}$$

(b) with the use of the microcells as shown:

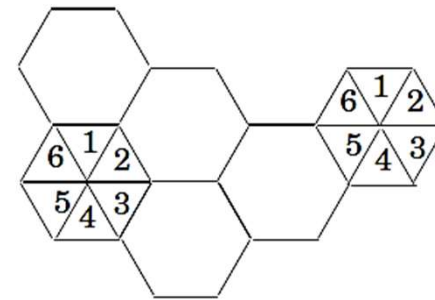
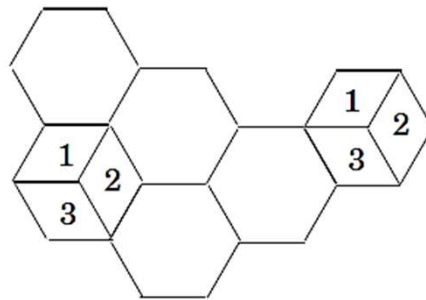
- The base station A is surrounded by six microcells.
- Therefore, the total number of base stations in the square area under consideration is equal to $5 + 6 = 11$.
- Since each base station has 60 channels, the total number of channels will be equal to $11 \times 60 = 660$ channels.
- This is a **2.2** times increase in capacity when compared to case (a).

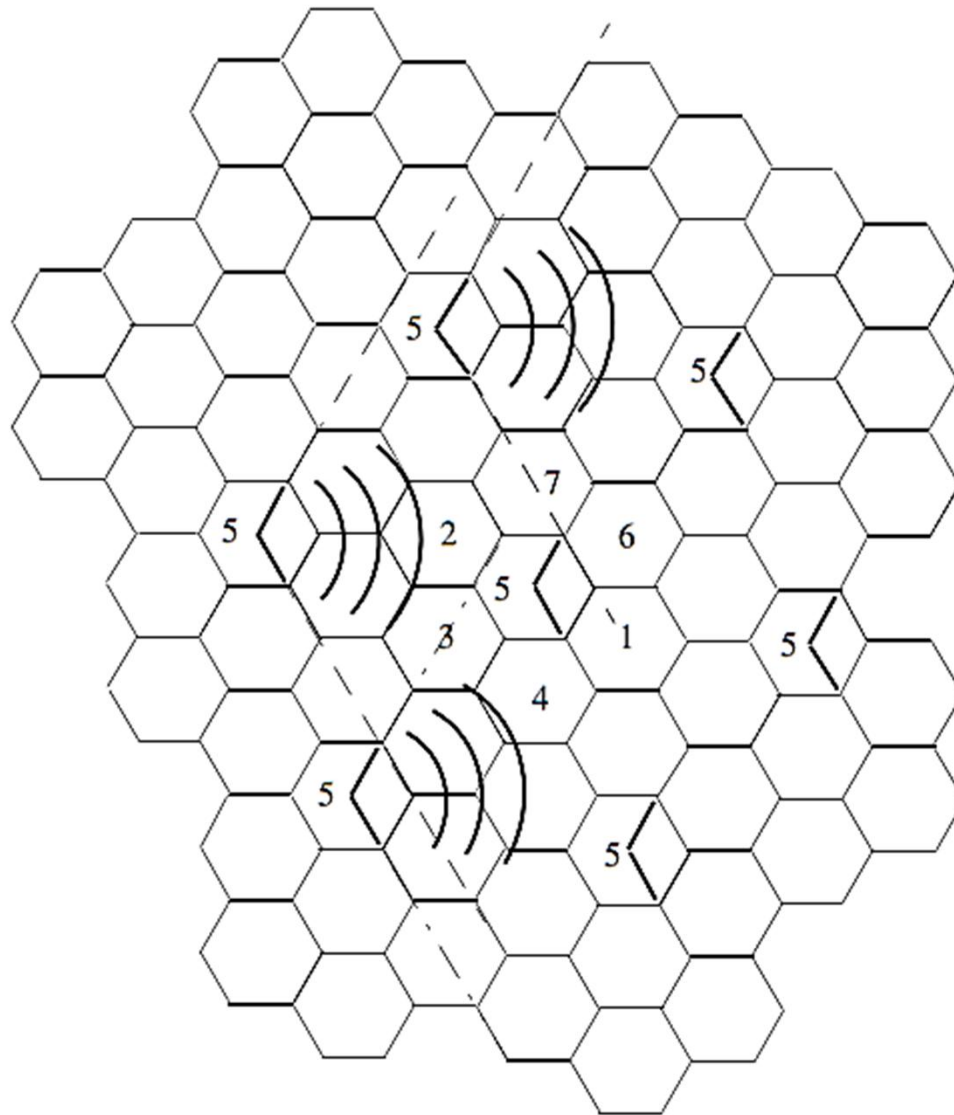
(c) if all the base stations are replaced by microcells:

- From the Figure, we see there are a total of $5 + 12 = 17$ base stations in the square region under study. Since each base station has 60 channels, the total number of channels will be equal to $17 \times 60 = 1020$ channels.
- This is a 3.4 times increase in capacity compared to case (a).
- Theoretically, if all cells were microcells having half the radius of the original cell, the capacity increase would approach four.

Sectoring

- Sectoring is a cellular **network optimization technique** used to divide a cell's coverage area into smaller, manageable sections or sectors.
- **Each sector has its own dedicated set of antennas** and equipment, allowing for more efficient use of resources and improved network performance.
- Sectoring is commonly employed in cellular network design **to enhance coverage, capacity, and quality of service.**





120° sectoring reduces interference from co-channel cells.

Out of the 6 co-channel cells in the first tier, only two of them interfere with the center cell.

If omnidirectional antennas were used at each base station, all six co-channel cells would interfere with the center cell.

Motivation:

- The primary motivation behind sectoring is to increase the overall capacity and efficiency of a cellular network.
- In a **traditional omni-directional cell** (a single cell with a single antenna pointing in all directions), all users within the cell share the available resources (frequency channels, power, and bandwidth).
- As the number of users grows, this can lead to congestion and reduced service quality.

Division of Coverage Area:

- Sectoring involves dividing a cell's coverage area into multiple smaller sections, typically using **directional antennas**.
- These sections are referred to as "sectors," and each sector is assigned its **own set of antennas, base station equipment, and frequency channels**.
- The number of sectors in a cell can vary, with common configurations being **three-sector and six-sector cells**.

Directional Antennas:

- Directional antennas are used in each sector to **focus the radio signal** in a specific direction.
- These antennas have a **narrower beam-width** compared to omni-directional antennas, which allows for precise control of where the signal is transmitted.
- Directional antennas help **reduce interference between neighboring sectors and cells**.

Benefits of Sectoring

- **Increased Capacity:** Reducing the number of users sharing the same resources in each sector. This leads to improved capacity and reduced congestion.
- **Enhanced Coverage:** Directional antennas can concentrate signal strength in the desired direction, providing better coverage where it's needed. This is particularly useful in areas with varying terrain and building structures.
- **Effective Frequency Reuse:** With directional antennas, the same frequencies can be reused in neighboring sectors without causing significant interference.

Benefits of Sectoring

- **Improved Quality of Service:** By reducing congestion and interference, sectoring improves the quality of voice and data services for users.
- **Enhanced Network Control:** Network operators can adjust the power levels and configurations of individual sectors to optimize network performance and meet specific demands.

Challenges and Considerations

- **Antenna Alignment:** Proper alignment of directional antennas is crucial to ensure optimal signal coverage and minimize interference. Precise planning and adjustment are required.
- **Handover Management:** Smoother handovers between sectors are essential to maintain call continuity as users move through different sectors. Seamless handover algorithms must be implemented.

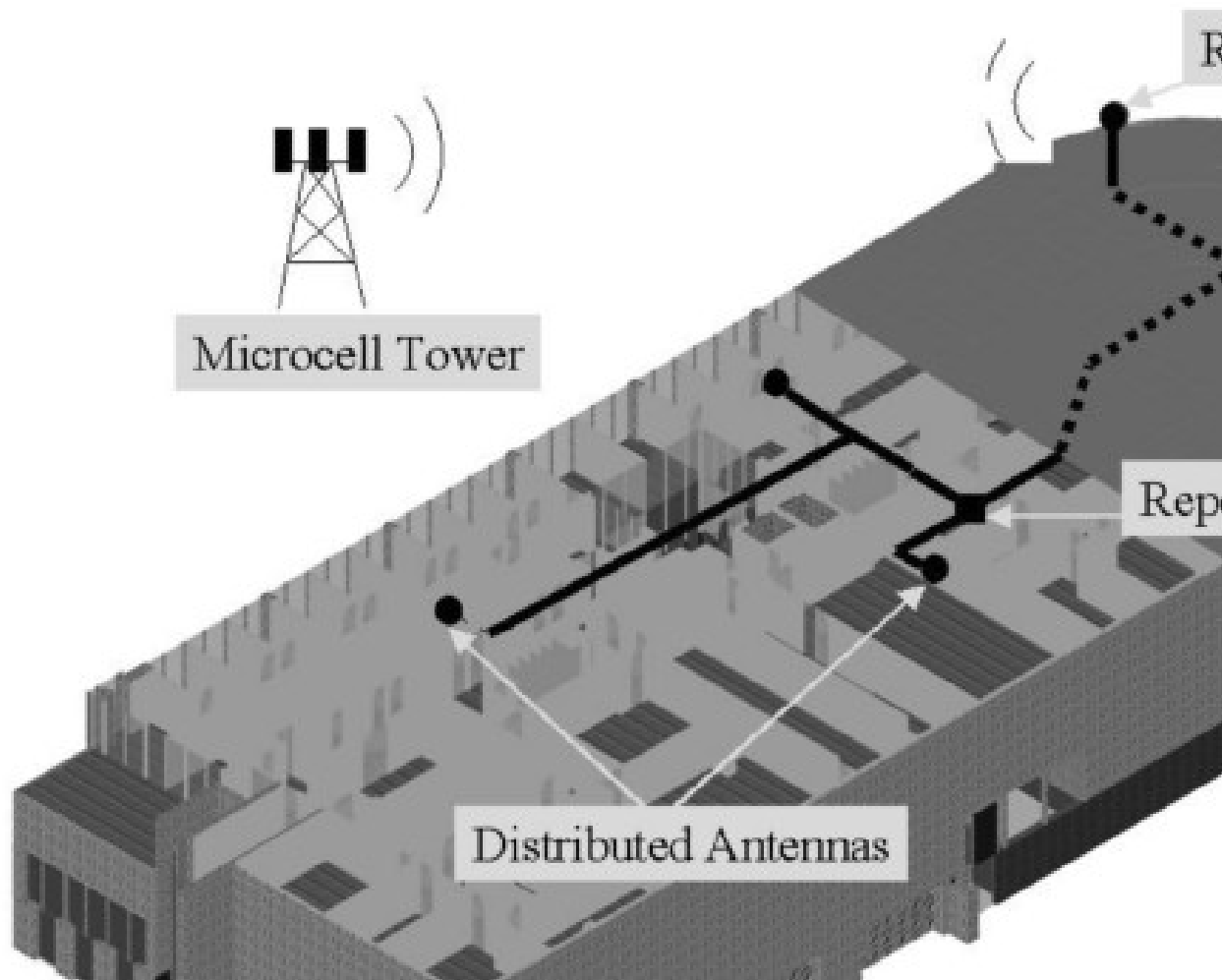
Challenges and Considerations

- **Increased Infrastructure Costs:** Sectoring may require additional base station equipment and antennas, leading to higher infrastructure costs.
- **Interference Management:** While sectoring reduces interference within the cell, careful interference management is still necessary, especially in areas with high user density.

Repeaters for Range Extension

- Often a wireless operator needs to provide dedicated coverage for hard-to-reach areas, such as within buildings, or in valleys or tunnels.
- Radio retransmitters, known as *repeaters*, are often used to provide such range extension capabilities.
- Repeaters are bidirectional in nature, and simultaneously send signals to and receive signals from a serving base station.
- Repeaters work using over-the-air signals, so they may be installed anywhere and are capable of repeating an entire cellular or PCS band.

Repeaters for Range Extension



Repeaters for Range Extension

- Upon receiving signals from a base station forward link, the repeater amplifies and reradiates the base station signals to the specific coverage region.
- Repeaters can be easily thought of as bidirectional “bent pipes” that retransmit what has been received.
- In practice, directional antennas or *distributed antenna systems (DAS)* are connected to the inputs or outputs of repeaters for localized spot coverage, particularly in tunnels or buildings.

Advantages

- By modifying the coverage of a serving cell, an operator is able to dedicate a certain amount of the base station's traffic for the areas covered by the repeater.
- Repeaters are increasingly being used to provide coverage into and around buildings, where coverage has been traditionally weak.
- They enhance connectivity, reduce dead zones, and ensure that users can access reliable wireless services across a broader range.

Limitation

- The received **noise and interference is also reradiated** by the repeater on both the forward and reverse link, so care must be taken to properly place the repeaters, and to adjust the various forward and reverse link amplifier levels and antenna patterns.
- Determining the proper location for repeaters and distributed antenna systems within buildings **requires careful planning**, particularly due to the fact that interference levels are reradiated into the building from the base station and from the interior of the building back to the base station.

Limitation

- Depending on the setup, repeaters can **introduce some latency** (delay) in data transmission.
- The coverage area of a repeater depends on factors like its power, antenna type, and environment. Coverage may vary, so **proper placement is critical**.
- Repeater **does not add capacity** to the system.

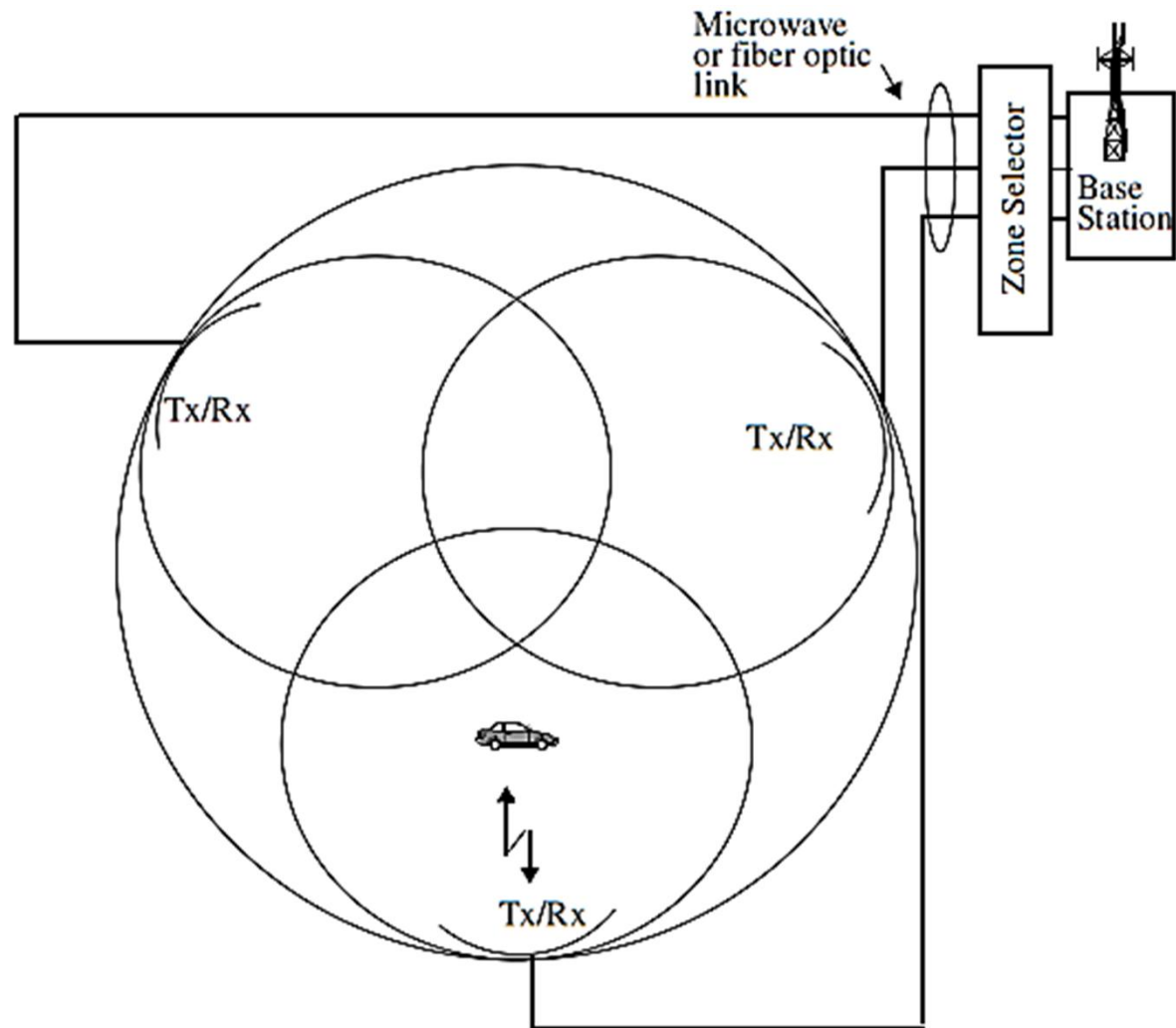
Microcell Zone Concept

- The increased number of handoffs required when sectoring is employed results in an increased load on the switching and control link elements of the mobile system.
- In a 7 cell reuse geometry, each of the three (or possibly more) zone sites are connected to a single base station and share the same radio equipment.
- Multiple zones and a single base station make up a cell. As a mobile travels within the cell, it is served by the zone with the strongest signal

Microcell Zone Concept

- This approach is superior to sectoring since **antennas are placed at the outer edges of the cell**, and any base station channel may be assigned to any zone by the base station.
- As a **mobile travels from one zone to another within the cell, it retains the same channel**. Thus, unlike in sectoring, a handoff is not required at the MSC when the mobile travels between zones within the cell.
- The base station simply **switches the channel to a different zone site**. In this way, a given channel is active only in the particular zone in which the mobile is traveling, and hence the base station radiation is localized and interference is reduced.

Microcell Zone



Benefits of Microcell Zones

- The channels are distributed in time and space by all three zones and are also reused in co-channel cells in the normal fashion.
- This technique is particularly useful along highways or along urban traffic corridors.
- While the cell maintains a particular coverage radius, the co-channel interference in the cellular system is reduced since a large central base station is replaced by several lower powered transmitters (zone transmitters) on the edges of the cell.
- Decreased co-channel interference improves the signal quality and also leads to an increase in capacity without the degradation in trunking efficiency caused by sectoring.

Limitations

- **Deployment Costs:** The deployment and maintenance can be costly due to the need for additional infrastructure, backhaul connections, and power supply.
- **Interference Management:** Given the high density of microcells within a zone, careful interference management is required to avoid signal interference and degradation of network performance.
- **Coordination with Macrocells:** Effective coordination between microcells within the zone and neighboring macrocells is essential to ensure seamless handovers as users move between coverage areas.